

# Chapter 7

## Semi-supervised Learning

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**Abstract.** In traditional supervised learning, one uses "labeled" data to build a model. However, labeling the training data for real-world applications is difficult, expensive, or time consuming, as it requires the effort of human annotators sometimes with specific domain experience and training. There are implicit costs associated with obtaining these labels from domain experts, such as limited time and financial resources. This is especially true for applications that involve learning with large number of class labels and sometimes with similarities among them. Semi-supervised learning (SSL) addresses this inherent bottleneck by allowing the model to integrate part or all of the available unlabeled data in its supervised learning. The goal is to maximize the learning performance of the model through such newly-labeled examples while minimizing the work required of human annotators. Exploiting unlabeled data to help improve the learning performance has become a hot topic during the last decade and it is divided into four main directions: SSL with graphs, SSL with generative models, semi-supervised support vector machines and SSL by disagreement (SSL with committees). This survey article provides an overview to research advances in this branch of machine learning.

### 1 Introduction

Supervised learning algorithms require a large amount of labeled training data in order to construct models with high prediction performance, see Figure 1. In many practical data mining applications such as computer-aided medical diagnosis [38], remote sensing image classification [49], speech recognition [32], email classification [33], or automated classification of text documents [44, 45], there is often an extremely inexpensive large pool of unlabeled data available. However, the data

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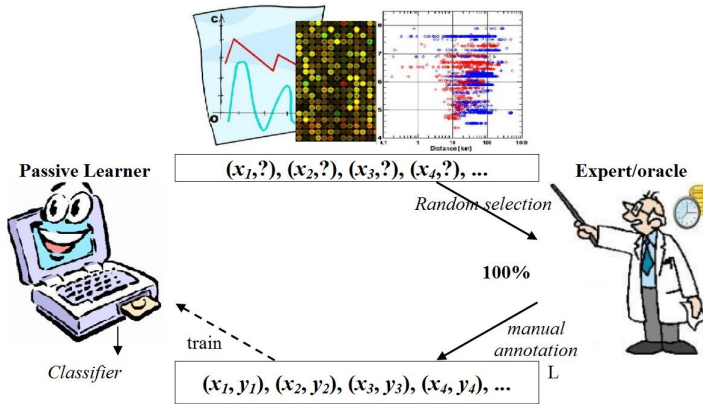
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**Fig. 1** Graphical illustration of traditional supervised learning

labeling process is often difficult, tedious, expensive, or time consuming, as it requires the efforts of human experts or special devices. Due to the difficulties in incorporating unlabeled data directly into traditional supervised learning algorithms such as support vector machines and RBF neural networks and the lack of a clear understanding of the value of unlabeled data in the learning process, the study of semi-supervised learning attracted attention only after the middle of 1990s. As the demand for automatic exploitation of unlabeled data increases, semi-supervised learning has become a hot topic.

In computer-aided diagnosis (CAD), mammography is a specific type of imaging that uses a low-dose x-ray system to examine breasts and it is used to aid in the early detection and diagnosis of breast diseases in women. There is a large number of mammographic images that can be obtained from routine examination but it is difficult to ask a physician or radiologist to search all images and highlight the abnormal areas of calcification that may indicate the presence of cancer. If we use supervised learning techniques to build a computer software to highlight these areas on the images, based on limited amount of diagnosed training images, it may be difficult to get an accurate diagnosis software. Then a question arises: can we exploit the abundant undiagnosed images [38] with the few diagnosed images to construct a more accurate software.

For remote sensing applications, the remote sensing sensors can produce data in large number of spectral bands. The objective of using such high resolution sensors is to discriminate among more ground cover classes and hence obtain a better understanding about the nature of the materials that cover the surface of the Earth. This large number of classes and large number of spectral bands require a large number of labeled training examples (pixels) from all the classes of interest. The class labels of such training examples are usually very expensive and time consuming to acquire [49]. The reason is that identifying the ground truth of the data must be gathered by visual inspection of the scene near the same time that the data is